

Age Effect on Pursuing MBA Decision

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Abstract

This research looks at the factors that influence students' choices to pursue an MBA and, in this framework, what effect their age might have on such a decision. Using these, we will apply, on a sample size of 10,000 observations, multiple regression models investigating the impact of the age variable and other control variables: work experience, pre-MBA salary, funding sources, expected post-MBA salary, and program delivery format (online vs. on campus). These results imply that the age factor does not have an insignificant and negligible effect on the choice to pursue an MBA. The same thing happens with the rest of the control variables. From the very low explanatory power of the models, with low statistical significance of predictors, it follows that the unobserved motivational, career or industry-specific factors could be much more important. The paper, therefore, recommends that future research incorporate variables driven by psychological and industry variables and adopt logistic regression models for binary outcomes to better capture decision-making processes.

Introduction

Research Question: Does age affect people's decision to pursue an MBA?

Specific Hypotheses:

- Null Hypothesis (H0): Age does not affect people's decision to pursue an MBA.
- Alternative Hypothesis (H1): Age affects people's decision to pursue an MBA.

Dependent Variable:

- Decided to Pursue an MBA? (Yes/No): This is a binary variable indicating whether an individual chooses to pursue an MBA.

Variable of Interest:

- Age: The primary independent variable representing the age of individuals at the time of the decision.

Control Variables: Based on surveying the literature, the following control variables are relevant as they may influence the decision to pursue an MBA and help isolate the effect of age:

- Years of Work Experience: Work experience can impact the decision to pursue an MBA, as individuals with more experience may seek an MBA to advance their careers or pivot to new roles.
- Annual Salary (Before MBA): Financial considerations are crucial in the decision-making process. Lower pre-MBA salaries might be a motivation for individuals to pursue an MBA for a higher salary.
- MBA Funding Source: The source of funding (self-funded, loan, scholarship, employer) can impact the affordability and attractiveness of pursuing an MBA.
- Expected Post-MBA Salary: The anticipated increase in salary after obtaining an MBA can motivate individuals to pursue the degree.
- Online vs. On-Campus MBA: Preferences for learning modes (online vs. on-campus) can influence the decision, especially for those who need flexibility.

Data

<https://www.kaggle.com/datasets/ashaychoudhary/dataset-mba-decision-after-bachelors>

The dataset "MBA Decision After Bachelor's" on Kaggle involves variables related to factors for making decisions regarding pursuing an MBA after completing a bachelor's degree. The dataset

provides a total of eighteen variables that may impact the decision to pursue an MBA. However, the analysis only includes six variables. Age is one of the primary factors that this paper will examine. The six variables' descriptive statistics are shown below and all variables are linear:

		complete _rate	character .min	character .max	character .empty	character .n_uniqu e	character .whitespa ce	numeric. mean
character	Decided to Pursue MBA	0	1	2	3	0	2	0
character	MBA Funding Source	0	1	4	11	0	4	0
character	Online vs On-Campus MBA	0	1	6	9	0	2	0

		complet e_rate	charact er.min	numeric .sd	numeric .p0	numeric .p25	numeric .p50	numeric .p75	numeric .p100	numeric . hist
numeric	Age	0	1	27.4943	4.03182 2549	21	24	27	31	34
numeric	Years of Work Experience	0	1	4.5335	2.86165 9774	0	2	5	7	9
numeric	Annual Salary Before MBA	0	1	75166.4 064	25850.0 706	30013	52862	74829.5	97273	119966
numeric	Expected Post MBA Salary	0	1	130321. 2289	40598.4 2199	60001	95320.5	130155. 5	165757	199999

	Age	Years of Work Experience	Annual Salary Before MBA	Expected.Post MBA Salary
Age	1	0	0.01	0.02
Years of Work Experience	0	1	-0.02	0
Annual Salary Before MBA	0.01	-0.02	1	0.02
Expected.Post MBA Salary	0.02	0	0.02	1

Analysis

Base Specification

The base regression examines age's role in the decision to pursue an MBA. The outcome variable is categorical, which presents the decision whether to pursue the MBA, the person decides (1) "Yes" or (0) "No". The base model is:

The only regressor this model controls is the effect of age on decisions to pursue an MBA. This base regression allows for an initial view of the influence of the independent variable, which in this case is age.

Alternative Specification:

To obtain more robust estimates and account for potential confounders, I included relevant control variables based on theoretical and empirical justifications:

- Years of Work Experience: Individuals with more experience may seek an MBA to advance or pivot in their careers.
- Annual Salary Before MBA: Lower pre-MBA salaries could motivate individuals to pursue an MBA for financial gains.
- Funding Source of MBA: Affordability issues, referring to different funding sources available for the respondents, affect the decision.
- Expected Post-MBA Salary: Higher expected post-MBA salaries will encourage people to obtain the degree.
- Online vs. On-Campus MBA: Preference for modes of learning affects decisions, particularly among those that need flexibility.

Control Variables Justifications

Including these controls helps to isolate the effect of age since there might be some confounding factors that could influence the decision to pursue an MBA and correlate with age. The robust standard error results obtained using vcovHC in R give more accurate and reliable estimates.

Results

	Dependent variable:						
	Decided to Pursue MBA						
	-1	-2	-3	-4	-5	-6	-7
Age	0.00004	0.0001					
	-0.001	-0.001					
Years of Work Experience	-0.001		-0.001				
	-0.002		-0.002				
Annual Salary Before MBA	0			0			
	0			0			
MBA Funding Source - Loan	0.009				0.009		
	-0.014				-0.014		
MBA Funding Source - Scholarship	-0.008				-0.008		
	-0.014				-0.014		
MBA Funding Source - Self-funded	-0.007				-0.007		
	-0.014				-0.014		
Expected Post MBA Salary	0					0	
	0					0	
Online vs On-Campus	0.001						0.001

	-0.01						-0.01
Constant	0.595***	0.589***	0.595***	0.592***	0.592***	0.590***	0.590***
	-0.042	-0.034	-0.009	-0.015	-0.01	-0.017	-0.007
Observations	10,000	10,000	10,000	10,000	10,000	10,000	10,000
R2	0.0002	0	0.00003	0	0.0002	0	0
Adjusted R2	-0.001	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001
Residual Std. Error	0.492 (df = 9991)	0.492 (df = 9998)	0.492 (df = 9998)	0.492 (df = 9998)	0.492 (df = 9996)	0.492 (df = 9998)	0.492 (df = 9998)
F Statistic	0.297 (df = 8; 9991)	0.004 (df = 1; 9998)	0.310 (df = 1; 9998)	0.006 (df = 1; 9998)	0.671 (df = 3; 9996)	0.005 (df = 1; 9998)	0.005 (df = 1; 9998)
Note:							
*p<0.1; **p<0.05; ***p<0.01							

The regression analysis yields several noteworthy findings regarding factors influencing MBA pursuit decisions. While the coefficients are statistically insignificant, their practical implications merit careful consideration:

- Age: It is a minimal coefficient (0.00004) and insignificant, which means age is not a significant factor in influencing one's choice for an MBA.
- Years of Work Experience: The coefficient, though small (-0.001), is insignificant, hence having a limited influence on the decision.
- Annual Salary Before MBA: This is also a very small negative coefficient, -0.00000, which is not significant enough to show a negligible impact.
- Sources of MBA financing: the coefficients are small and insignificant, showing that the source of funding is not decisive in choosing between Loan, Scholarship, or Self-funded.
- Expected Salary after MBA: From the coefficient value 0.000, no significant effect was deduced on the decision.
- Online vs. On Campus MBA: The positive coefficient is small, 0.001, and hence insignificant, showing that the mode of study, online or on campus, is not of much importance.

Considering the size of the coefficients and their nonsignificance, it is possible to infer that these predictors have no relevant quantitative effect on the decision of an MBA application. The conducted regression does not answer which factors significantly contribute to the MBA pursuit decision. All predictors demonstrate negligible and non-significant effects, suggesting the influence of unobserved factors that may play a more crucial role in making such a decision. The table presents seven different model specifications, allowing us to assess robustness:

- Model 1: Full model with all variables
- Models 2-7: Individual variable specifications

Given the general low explanatory power of the model, changes in the specification-as captured by the different regression columns-make little difference to the coefficients or their statistical significance. This would, in turn, suggest that the addition or deletion of individual variables does little to improve the model fit of the outcome.

Sample Size and Representativeness: The sample size in this dataset is 10,000; this can make a decent generalization. Given that the information on diversity, such as how representative it was in geographical, demographic, and professional perspectives, is unknown, it remains questionable how generalization to wider groups would come into place. The MBA decisions could be influenced by regional economic factors or educational trends, which might not be generalizable to other contexts or countries.

Conclusion

This study tries to determine whether age is a factor in the decision to pursue an MBA. The results indicate that age, together with other demographic and economic factors such as work experience, pre-MBA salary, and expected post-MBA salary, does not have a statistically significant effect on this decision. The low R^2 values and insignificant coefficients across different specifications suggest that traditional variables may not capture the key drivers of MBA

pursuit. More crucially, these factors are unobserved: personal motivations, career aspirations, and industry trends.

Research in the future should be much more broadly gauged with the inclusion of psychological and career-related variables. Logistic regression models would also afford a better conceptual framework for understanding the binary outcomes of MBA decisions. This study highlights avenues other than those that have so far been followed in research variables and areas of methodological improvement that promise better insights into educational decision-making.